Tianyi Huang

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Education

National University of Singapore — **Jul 2025 – Present**
Ph.D. in Computing — Advisor: Prof. Bohan Wang — Singapore

University of Illinois Urbana-Champaign — **Aug 2021 – May 2025**
B.S. in Computer Science — GPA: 3.97/4.0, Highest Honors — Champaign, IL

Experience

University of Virginia — **Aug 2024 – Aug 2025**
Visiting Researcher (Advisor: Prof. Wenxi Wang) — Remote

- Conducted research on enhancing the efficiency of CDCL based SAT-solving algorithms
- Built a parallelized generator for creating **10K** artificial SAT formulas and corresponding backdoors across six categories, including k-SAT, graph coloring, and Tseitin formulas

University of Illinois Urbana-Champaign — **May 2024 – Dec 2024**
Undergraduate Research Assistant — Champaign, IL

- Participated in the *SysNet* group, analyzing over **50 bugs** in widely-used Kubernetes operators to identify root causes, categorize failure modes, and propose fixes
- Developed a **CodeQL static analysis tool** to detect interface-related implementation bugs in Kubernetes operators

University of Illinois at Urbana-Champaign — **Aug 2022 – Dec 2022**
Course Assistant — Champaign, IL

- Served as a course assistant for *MATH 257: Linear Algebra with Computational Applications*, conducting weekly office hours to assist and resolve inquiries for over 100 students
- Delivered hands-on guidance to students during weekly lab sessions, assisting with coding exercises

Publications

Shellular Metamaterial Design via Compact Electric Potential Parametrization. **Tianyi Huang**, Chang Liu, Bohan Wang. *SIGGRAPH*, 2026

Projects

QSync: High-performance Multi-layered Audio Synthesizer | C, WebAssembly

- Implemented a high-performance real-time audio synthesizer in pure C supporting stereo output, low-latency playback down to approximately **10*ms*** under tuned buffer settings; users can customize instruments through sinusoidal oscillator presets, filters, phase offsets, and amplitude controls, and define custom effect pedals through a simple interface-based architecture. The system supports cross-platform deployment, including web applications through Wasm

Octave: Music Programming Language | C, Zig, WebAssembly, MIDI

- Research topic advised by Prof. Elsa L. Gunter. Designed and implemented a Turing-complete, notation-inspired music programming language from scratch in C, with expressive syntax for composition workflows, variables, control flow, concurrent tracks, and MIDI output
- Implemented the compiler backend and contributed to frontend components, including grammar design, lexing, parsing, and type checking. Enabled browser-based execution via Wasm with an interactive web demo

EmuEngine: Serverless Blog Engine | TypeScript, SQL, Tailwind CSS, Preact

- Implemented a fully serverless blog engine backed by Cloudflare Workers, D1, and R2 for low-cost, fast, and secure personal blogging; a live deployment is available at my blog. Frontend driven by Tailwind CSS
- Supported full blog functionality with additional production-oriented features, including one-click database hot backup, comment moderation, online editing, and media asset deduplication

Technical Skills

Languages: C/C++, CUDA, Python, OCaml, Zig, WebAssembly, SQL, JavaScript/TypeScript, HTML/CSS
Graphics & Geometry: Vulkan, Blender, OpenVDB
AI/ML: PyTorch, Matplotlib, scikit-learn, OpenCV
Other: Conda, Adobe Illustrator, LaTeX, CodeQL, Vim, Kubernetes